

Introduction to Contract Theory and Auctions
Bayesian Auctions: Symmetric, Independent, Private Values

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The symmetric model with independent private values

We will now study a class of models that satisfy the following assumptions:

- A1 Single object for sale and the object is indivisible.
- A2 The seller is willing to sell the object at any nonnegative price.
- A3 There are n buyers.
- A4 Private values and symmetric values: All buyers have the same set of possible private values $\mathcal{V}$. This set can be a closed bounded interval $[0, \bar{v}]$ or the set of nonnegative numbers $[0, \infty)$. Every buyer knows her private value for the object. The random values $V_1, V_2, \dots, V_n$ of the private values of the buyers are independent and identically distributed. Denote by F the common cumulative distribution function of the random variables V_i , $i = 1, \dots, n$. The support of this distribution is $\mathcal{V}$. The joint distribution is then $F^n = F \times \dots \times F$.
- A5 Continuity: For each i , the random variable V_i is continuous, and its density function, denoted by f , is continuous and positive.
- A6 Risk neutrality: All buyers are risk neutral, and therefore seek to maximize their expected profits.

A1–A6 are common knowledge among the buyers. Any auction satisfying A1–A6 is called a **symmetric auction with independent private values (SIPV)**.

Definition 6.1

In a symmetric auction with independent private values, an equilibrium profile $(\beta_1^*, \beta_2^*, \dots, \beta_n^*)$ is called a **symmetric equilibrium** if $\beta_i^* = \beta_j^*$ for all $i, j \leq n$; that is, all buyers implement the same strategy.

When the profile $\beta^* = (\beta_i^*)_{i \in N}$ is a symmetric equilibrium, we *abuse notation* and denote the common strategy also as β^* , i.e., $\beta^* = \beta_i^*$ for every $i \in N$. Such a strategy is called a **symmetric equilibrium strategy**.

Definition 6.2

A symmetric equilibrium strategy β^* is monotonically increasing if the higher the private value, the higher the buyer's bid:

$$v < v' \Rightarrow \beta^*(v) < \beta^*(v') \quad \forall v, v' \in \mathcal{V}.$$

Example: Two bidders with i.i.d Uniform valuations

Example 6.3

Suppose that V_1 and V_2 are independent and identically distributed (i.i.d.) Uniform random variables over the interval $[0, 1]$, i.e., $V_1, V_2 \sim U[0, 1]$. In this example, we will show that in a sealed-bid first-price auction, the following strategy is a symmetric equilibrium:

$$\beta_i^*(v_i) = \frac{v_i}{2} \quad i = 1, 2.$$

See pages below for my handwritten notes of the derivation of this example, which I did in the whiteboard during class.

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Two buyers with uniformly distributed values.

Suppose there are two buyers, and that $V_i \sim U[0,1]$ for $i=1,2$, and that $V_1 \perp V_2$.

Cheating: Consider $\beta_i^*(v_i) = \frac{v_i}{2}$ $i=1,2$, we will show that this is a symmetric eq. strategy.

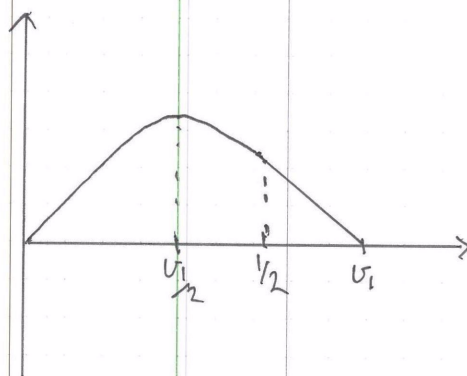
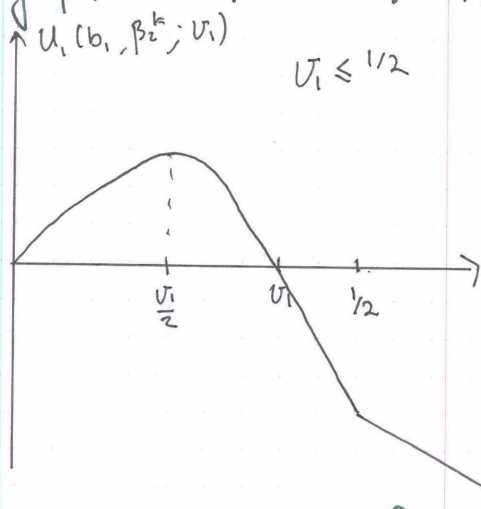
Uniform CDF: $F = \begin{cases} 0 & \text{if } t < 0 \\ t & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } t > 1 \end{cases}$

Suppose that buyer 2 implements this strategy. Then if buyer's 1 private value is v_1 , and her submitted bid is b_1 , her expected payoff is:

$$\begin{aligned} u_1(b_1, \beta_2^*; v_1) &= u_1(b_1, \frac{v_2}{2}; v_1) \\ &= \Pr(b_1 > \frac{v_2}{2}) (v_1 - b_1) \\ &= \Pr(2b_1 > v_2) (v_1 - b_1) \\ &= (1 - F(2b_1)) (v_1 - b_1) \\ &= \begin{cases} (1 - 2b_1) (v_1 - b_1) & \text{if } 0 \leq 2b_1 \leq 1 \\ v_1 - b_1 & \text{if } 2b_1 \geq 1 \end{cases} \\ &= \min\{2b_1, 1\} (v_1 - b_1) \end{aligned}$$

This function is quadratic on $b_1 \in [0, 1/2]$ (with a maximum at $b_1 = \frac{v_1}{2}$), and linear with negative slope when $b_1 \geq 1/2$.

The graph $b_1 \mapsto u_1(b_1; \beta_2^*; v_1)$ is shown below:



In both cases, the function attains a maximum at the point $b_1 = \frac{v_1}{2} \Rightarrow b_1^*(v_1) = \frac{v_1}{2}$ is a best response to β_2^* .

4)

which means that $\beta^* = (\beta_1^*, \beta_2^*)$ is a symmetric equilibrium.

• In the SFPA FPSA the symmetric equilibrium is given by

$$\beta_i^*(v_i) = \frac{v_i}{2}$$

• In the SPFA, a symmetric eq is given by $\beta_i^*(v_i) = v_i$

Which auction method is preferable to the seller? We now turn to the calculation of the expected revenue for the seller in both auction formats. The expected revenue equals the expected sale price:

• For the first-price auction this is

$$E[\max\{\frac{V_1}{2}, \frac{V_2}{2}\}] = \frac{1}{2} E[\max\{V_1, V_2\}]$$

Denote by $Z = \max\{V_1, V_2\}$. We need to compute the CDF of this random variable: since $V_1 \perp V_2$ and they are both uniform $U[0, 1]$ we have:

$$\begin{aligned} F_Z(z) &= \Pr(Z \leq z) = \Pr(\max\{V_1, V_2\} \leq z) = \Pr(V_1 \leq z, V_2 \leq z) \\ &= \Pr(V_1 \leq z) \times \Pr(V_2 \leq z) = z^2 \end{aligned}$$

Thus the density is:

$$f_Z(z) = \begin{cases} 2z & 0 \leq z \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} \text{Thus } \frac{1}{2} E(Z) &= \frac{1}{2} \int_0^1 z f_Z(z) dz = \frac{1}{2} \int_0^1 2z^2 dz = \int_0^1 z^2 dz \\ &= \left. \frac{z^3}{3} \right|_0^1 = \frac{1}{3} \end{aligned}$$

The seller's revenue in the second-price auction is $E[\min\{Z, Z\}] = E[\min\{V_1, V_2\}]$. Note that: (trick)

$\min\{V_1, V_2\} + \max\{V_1, V_2\} = V_1 + V_2$. Thus:

$$E[\min\{V_1, V_2\}] + E[\max\{V_1, V_2\}] = E[V_1] + E[V_2] = \frac{1}{2} + \frac{1}{2} = 1$$

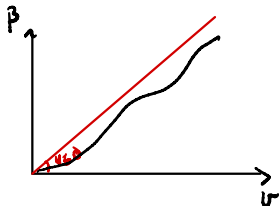
$$\Rightarrow E[\min\{V_1, V_2\}] = \overset{= 2/3}{1 - 1/3} = \frac{1}{3} //$$

Heuristic derivation of the symmetric eq. strategy with n bidders in the SIPV First-Price Auction.

Suppose that bidders $j \neq 1$ follow the symmetric, monotonically increasing, and differentiable equilibrium strategy β . Suppose that bidder 1 has value $V_1 = v_1$ and bids b .

First note: we cannot have $b > \beta(x)$ for any x . Why? Player 1 wins but pays b , needs to pay b . Player can increase her payoff by bidding slightly less, and still win. $\Rightarrow$ We only need to consider $b \leq \beta(x)$.

Secondly: bidder 1 with value $V_1 = 0$ must submit a bid of zero, $\beta(0) = 0$ (you should bid the lowest amount if your valuation is the lowest possible).



Bidder 1 wins: $b > \max_{j \neq 1} \beta(V_j)$

since β is increasing $\max_{j \neq 1} \beta(V_j) = \beta(\max_{j \neq 1} V_j)$

Let $Y = \max_{j \neq 1} V_j$ note: Y is a RV $= \beta(Y)$

↳ CDF of Y ? $\Pr(Y \leq y) = \Pr(\max_{j \neq 1} V_j \leq y) = \Pr(V_2 \leq y, V_3 \leq y, \dots, V_n \leq y)$

V_j are i.i.d. $= \prod_{j \neq 1} \Pr(V_j \leq y) = F(y)^{n-1} \equiv G(y)$

$Y = \left(\max_{j \neq 1} V_j \right) \sim G(y) \Rightarrow$ density of G g

bidder 1 wins when $b > \beta(Y) \iff \beta^{-1}(b) > Y$

Expected Payoff: $\Pr(\text{Winning}) \times (v_1 - b) = \Pr(Y < \beta^{-1}(b)) (v_1 - b)$
 $= G(\beta^{-1}(b)) (v_1 - b)$

We will differentiate the expression $G(\beta^{-1}(b)) (v_1 - b)$ wrt. b

Detour. Inverse function theorem: If f is a continuously differentiable function with non-zero derivative at the point a , then if f is injective in a neighborhood of a , the inverse is continuously differentiable near $b = f(a)$ and the derivative of the inverse function at b is:

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}$$

$$\frac{d}{db} (G(\beta^{-1}(b))(v_1 - b)) = \frac{g(\beta^{-1}(b))}{\beta'(\beta^{-1}(b))} (v_1 - b) - G(\beta^{-1}(b)) \stackrel{\text{FOC}}{=} 0$$

At a symmetric equilibrium: $b = \beta(v_1) \Rightarrow \beta^{-1}(b) = v_1$

$$\frac{g(v_1)}{\beta'(v_1)} \cdot (v_1 - \beta(v_1)) - G(v_1) = 0$$

$$\frac{g(u_1)}{\beta'(u_1)} \cdot (u_1 - \beta(u_1)) - G(u_1) = 0 \Rightarrow g(u_1)(u_1 - \beta(u_1)) - G(u_1)\beta'(u_1) = 0$$

$$\Leftrightarrow G(u_1)\beta'(u_1) = g(u_1)(u_1 - \beta(u_1))$$

$$\Leftrightarrow G(u_1)\beta'(u_1) + g(u_1)\beta(u_1) = g(u_1)u_1, \quad \beta(0) = 0$$

$$\Leftrightarrow \frac{d}{du_1} [G(u_1)\beta(u_1)] = g(u_1)u_1 \quad ; \quad \beta(0) = 0 \quad \text{we have IVP}$$

(initial value problem
diff eq. + boundary cond.)

$$\int_0^{u_1} \frac{d}{dt} [G(t)\beta(t)] dt = \int_0^{u_1} g(y)y dy$$

$$G(t)\beta(t) \Big|_0^{u_1} = \int_0^{u_1} g(y)y dy$$

$$G(u_1)\beta(u_1) - \underbrace{G(0)\beta(0)}_0 = \int_0^{u_1} g(y)y dy \Rightarrow \beta(u_1) = \frac{1}{G(u_1)} \int_0^{u_1} y g(y) dy$$

$$\therefore \boxed{\beta(u_1) = E[Y | Y \leq u_1]}$$

$$= \frac{1}{\Pr(Y \leq u_1)} \int_0^{u_1} y g(y) dy$$

Heuristic derivation of the symmetric eq. strategy with n bidders in the SIPV First-Price Auction.

Suppose that bidders $j \neq 1$ follow the symmetric, monotonically increasing, and differentiable equilibrium strategy β . Suppose that bidder 1 has value $V_1 = v_1$ and bids b .

$$\begin{aligned}\beta(v_i) &= E[Y | Y \leq v] \quad ; \quad Y = \max_{j \neq i} V_j \sim G(y) \text{ ; pdf } g(y) \\ &= \frac{1}{G(v)} \int_0^v y g(y) dy \quad \quad \quad = Y_1^{(n-1)} \sim G_1^{(n-1)}(y) \text{ ; } g_1^{(n-1)}(y)\end{aligned}$$

added on Thursday 3/11/22

Theorem 6.4

Let $Y = \max_{j \neq i} \{V_j\}$ be the random variable whose value equals the highest private value of all buyers j other than i , and let G and g denote respectively the CDF and PDF of Y . In a symmetric first-price auction with independent private values the following strategy defines a monotonically increasing, continuous, differentiable, symmetric equilibrium:

$$\beta(v) = E[Y|Y \leq v], \quad \forall v \in \mathcal{V} \setminus \{0\}$$

$\downarrow$
 $[0, \bar{v})$
 $[0, \infty)$

and $\beta(0) = 0$.

Proof.

Step 1: $\beta(v)$ is monotonically increasing.

Remark: For any R.V. X and any pair of events such that $A \cap B = \emptyset$

$$E[X|A \cup B] = P(A|A \cup B) E[X|A] + P(B|A \cup B) E[X|B]$$

Note that $g > 0$ for all y .

Let v be an interior point of $\mathcal{V} = [0, \bar{v}]$, $\Rightarrow v > 0, v < \bar{v}$.

Then for any $\delta > 0$ such that $v + \delta < \bar{v}$ we want to show that

$$\beta(v + \delta) > \beta(v) \quad \text{Goal!}$$

$$\beta(v + \delta) = E[Y | Y \leq v + \delta] \quad \text{We decompose } Y \leq v + \delta \text{ as two events}$$

$$Y \leq v \quad \text{AND} \quad v < Y \leq v + \delta$$

$$= \Pr(Y \leq v | Y \leq v + \delta) E[Y | Y \leq v]$$

$$+ \Pr(v < Y \leq v + \delta | Y \leq v + \delta) E[Y | v < Y \leq v + \delta]$$

$$= \frac{\Pr(Y \leq v)}{\Pr(Y \leq v + \delta)} E[Y | Y \leq v] + \frac{\Pr(v < Y \leq v + \delta)}{\Pr(Y \leq v + \delta)} E[Y | v < Y \leq v + \delta]$$

Since g is positive: $0 < E[Y | Y \leq v] \leq v < E[Y | v < Y \leq v + \delta]$ because $v > 0$

$\beta(v + \delta)$ is a weighted average of positive terms and $\beta(v) \leq v$

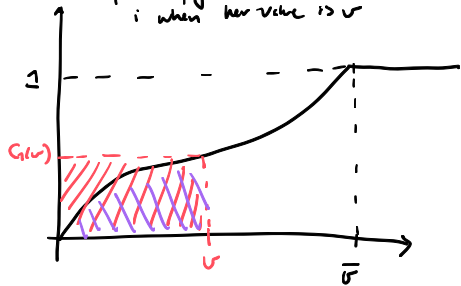
$$\Rightarrow \beta(v + \delta) > \beta(v)$$

Note that
$$\beta(v) G(v) = \int_0^v y g(y) dy = y G(y) \Big|_0^v - \int_0^v G(y) dy = v G(v) - \int_0^v G(y) dy$$

$$u = y \quad s = G(y)$$

$$du = dy \quad ds = g(y) dy$$

$$\therefore \underbrace{\beta(v) G(v)}_{\text{expected payment of player } i \text{ when her value is } v} = v G(v) - \underbrace{\int_0^v G(y) dy}_{\text{degree of value shading in FPA}}$$



$G(v)$

$$\boxed{\beta(v) = v - \int_0^v \frac{G(y)}{G(v)} dy}$$

Proof (Cont.)

Step 2: $\beta(v)$ is continuous.

Continuity at $v=0$, For each $v>0$: $0 \leq \beta(v) = E[Y|Y \leq v] \leq v$.

Taking the limit $v \rightarrow 0 \Rightarrow 0 \leq \beta(0) \leq 0 \Rightarrow \beta(0) = 0$.

Continuity at interior points of $\mathcal{V}$

$\beta(v) = \frac{\int_0^v y g(y) dy}{G(v)}$, we have a quotient of two continuous functions and $G(v) \neq 0$ in $v \in (0, \bar{v}) \Rightarrow \beta(v)$ is continuous

Step 3: $\beta(v)$ is differentiable. $\beta(v) = v - \int_0^v \frac{G(y)}{G(v)} dy$

$$\beta'(v) = 1 - \frac{G^2(v) - g(v) \int_0^v G(y) dy}{G^2(v)} = \left(\frac{g(v)}{G(v)} \right) \left(\int_0^v \frac{G(y)}{G(v)} dy \right)$$

Actually β is continuously differentiable.

Proof (Cont.)

Step 4: $\beta(v)$ is a symmetric equilibrium. Suppose that all bidders other than i follow β .
Goal: show that it is a best response for i to bid according to β given her value v_i .

Let v_i be bidder i 's private value. If $\mathcal{V} = [0, \bar{v}]$, suppose that $v_i < \bar{v}$, which occurs with probability 1.

Since β is increasing & continuous $\Rightarrow \beta^{-1}$ exists & is also continuous. Bidder i 's expected profit is:

$$U_i(b_i, \beta_{-i}; v_i) = \Pr(\beta(Y) < b_i) (v_i - b_i) \quad \text{where } Y = \max_{j \neq i} \{V_j\}$$

Observations:

- If bidder i bids $b_i = 0$, with prob. 1 she doesn't win.
Why? The density of Y has support on $\mathcal{V}$ so that $\Pr(Y > 0) = 1$
and since β is monotonically increasing, with prob. 1 some bidder bids more than zero $\Rightarrow U_i(0, \beta_{-i}; v_i) = 0$.
- If bidder i bids $b_i > v_i$, her payoff becomes non-positive:
 $U_i(b_i, \beta_{-i}; v_i) \leq 0 \quad \forall b_i > v_i$

• Since V_i has support on $\mathcal{V}$, $\Pr(\beta(Y) < b_i)$ is positive for all $b_i > 0$. Thus, in the interval $(0, v_i)$ the function $U_i(b_i, \beta_{-i}; v_i)$ is the product of two positive functions $\Rightarrow U_i(b_i, \beta_{-i}; v_i) > 0 \quad \forall b_i \in (0, v_i)$ and $U_i(b_i, \beta_{-i}; v_i) \leq 0 \quad \forall b_i \notin (0, v_i)$
 $\therefore U_i(b_i, \beta_{-i}; v_i)$ must attain a maximum in $(0, v_i)$.



Now we rewrite the payoff U_i in a useful way:

$$\begin{aligned} U_i(b_i, \beta_{-i}; v_i) &= P_r(\beta(Y) < b_i)(v_i - b_i) \\ &= P_r(Y < \beta^{-1}(b_i))(v_i - b_i) \\ &= G(\beta^{-1}(b_i))(v_i - \beta(\beta^{-1}(b_i))) \\ &= G(\beta^{-1}(b_i))(v_i - E[Y | Y < \beta^{-1}(b_i)]) \end{aligned}$$

Let b_i be an interior point of $\beta(\mathcal{V})$ which is the image of $\mathcal{V}$ under β , i.e.
 $\beta(\mathcal{V}) = \{b \in \mathbb{R} : \exists v \in \mathcal{V} \text{ such that } \beta(v) = b\}$; denote $z_i = \beta^{-1}(b_i)$, where $z_i \in \mathcal{V}$

$$\Rightarrow \beta(z_i) = b_i.$$

$$\text{Hence: } U_i(\beta(z_i); \beta_{-i}; v_i) = \underbrace{G(z_i)(v_i - E[Y | Y \leq z_i])}_{\text{call this side } h(z_i)}$$

Recall that we derived:

$$G(v) \cdot \underbrace{E[Y | Y \leq v]}_{\beta(v)} = vG(v) - \int_0^v G(y) dy$$

$$\begin{aligned} \text{Now } h(z_i) &= G(z_i)(v_i - E[Y | Y \leq z_i]) \\ &= G(z_i)(v_i - z_i) + G(z_i)z_i - G(z_i)E[Y | Y \leq z_i] \\ &= G(z_i)(v_i - z_i) + G(z_i)z_i - z_i G(z_i) + \int_0^{z_i} G(y) dy \\ &= G(z_i)(v_i - z_i) + \int_0^{z_i} G(y) dy \end{aligned}$$

Now we can use the F.O.C on $h(z_i)$

$$h(z_i) = g(z_i)(v_i - z_i) - G(z_i) + G(z_i) = 0$$

$$\Leftrightarrow g(z_i)(v_i - z_i) = 0$$

Since $g > 0 \Rightarrow z_i^* = v_i$, i.e., the optimum is achieved when $z_i^* = v_i$
 $\Rightarrow$ Player i 's best reply, when all others implement β , is to
bid $\beta^{-1}(b_i) = z_i^* = v_i \Rightarrow b_i^* = \beta(v_i)$ when her value is v_i 

Theorem 6.5

In the symmetric equilibrium given by Theorem 6.4, the expected payment that a buyer with private value v makes for the object is $G_Y(v) \times E[Y|Y \leq v]$ in both sealed-bid first-price and second-price auctions.

Proof.

At the eq. of the FPA $\beta^F(v) = E(Y|Y \leq v) \Rightarrow$ The expected payment is $G(v)E(Y|Y \leq v)$

At the eq. of the SPA $\beta^S(v) = v$, buyer with value v wins with prob. $G(v)$ and pays $E(Y|Y \leq v) \Rightarrow$ the expected amount she pays is $G(v)E(Y|Y \leq v)$



Theorem 6.6

In a symmetric sealed-bid auction with independent private values, at the symmetric equilibrium, the expected revenue of a seller in both sealed-bid first-price and second-price auctions, is

$$\pi = n \int_{v \in V} G_Y(v) E[Y|Y \leq v] f(v) dv$$

Proof.

The expected payment of a buyer with value v is $G_Y(v) E[Y|Y \leq v]$

$\Rightarrow$ The expected payment made by each buyer is

$$\int_{v \in V} G_Y(v) E[Y|Y \leq v] f(v) dv$$

Since the seller expected revenue is the sum of this for all other n buyers, the statement above follows

